

IZELMADENE ACHRAF

Engineering Student - Electrical & Power Engineering

achraf.izelmadene@gmail.com +212699098736 in/ achraf-izelmadene Marrakech, Morocco

PROFILE

Seeking a 6-month final-year internship starting February 1st, 2027, to leverage my skills in MV/LV sizing and photovoltaic engineering. Passionate about energy transition, I aim to contribute to the design of photovoltaic power plants and the sizing of electrical installations.

EXPERIENCE

Design Office Intern

Core Engineering Solutions - Berrechid, Casablanca Jul. - Aug. 2026

- Sizing of an MV/LV substation and integration of a self-consumption solar PV installation.
- Creation of single-line diagrams, power balances, and calculation notes (AutoCAD, Caneco BT, PVsyst).

Industrial Production Intern

Menara Prefa - Marrakech Jul. 2024

- Migration of an industrial PLC to Siemens S7 using TIA Portal, followed by preventive/corrective maintenance.
- Diagnosis of electrical faults and participation in the continuous improvement of industrial processes.

PROJECTS

Precision Agriculture IoT Ecosystem

- Deployment of a LoRa wireless IoT network connecting sensor nodes (Arduino) to a gateway (ESP32).
- Integration of an Edge AI (TFLite) and development of a web dashboard for global system supervision.

Predictive Maintenance 4.0 via AutoML

- Comparison of AutoML tools TPOT vs. Auto-WEKA on IoT data (vibration, temperature).
- Time-series cross-validation, SMOTE, automated ML pipelines, and NDT study via oil analysis.

Control of an AC Voltage Controller via Arduino

- Precise modeling in Simulink and Hardware-in-the-Loop (HIL) interfacing with an Arduino board.
- Real-time control and regulation of electrical power via firing angle control.

EDUCATION

Engineering Student in Electrical Engineering and Smart Systems

Moroccan School of Engineering Sciences (EMSI) - Marrakech 2025 - Present

Electrical Networks and Installations, Electrical Safety, Power Electronics, Supervision & Digital Twin, Project Management...

Bachelor in Physical Sciences, Control Systems option

Semlalia Faculty of Sciences - Marrakech 2020 - 2024

Electromagnetism, Electronics, Signal Processing, Control Systems, Instrumentation, Industrial IT...

SKILLS

Engineering & Tools : Caneco BT, PVsyst, AutoCAD, MATLAB, Simulink, 3DEXPERIENCE, Git, Docker
Programming & AI : Python, C/C++, JavaScript, HTML/CSS, Machine Learning, Scikit-learn, AutoML, MLOps
Soft Skills : Analytical Rigor, Technical Curiosity, Autonomy, Synthesis Skills, Teamwork

CERTIFICATIONS

- **Foundations of IoT Systems and Industrial Automation** — LearnQuest (Coursera)
- **Sensors and Sensor Circuit Design** — University of Colorado Boulder (Coursera)
- **Certified 3DEXPERIENCE Collaborative Industry Innovator** — Dassault Systèmes
- **MATLAB / Simulink Onramp** — MathWorks

ACTIVITIES & COMPETITIONS

- **1st Prize** -- Hack The Intelligence (AISEC Summit 2026)
- **2nd Prize** -- Vision AI Competition (EMSI 2026)
- **Scientific Presentation** -- COCIA'26 Congress (Casablanca)
- **Participation** -- Hackathon RallyIA (2026)

LANGUAGES & MISCELLANEOUS

Arabic : Native **French :** DELF B2 **English :** EFSET B2
Driving License : B **Hobbies :** Reading, Chess, DIY Electronics